A Suboptimality Approach to Distributed Linear Quadratic Optimal Control

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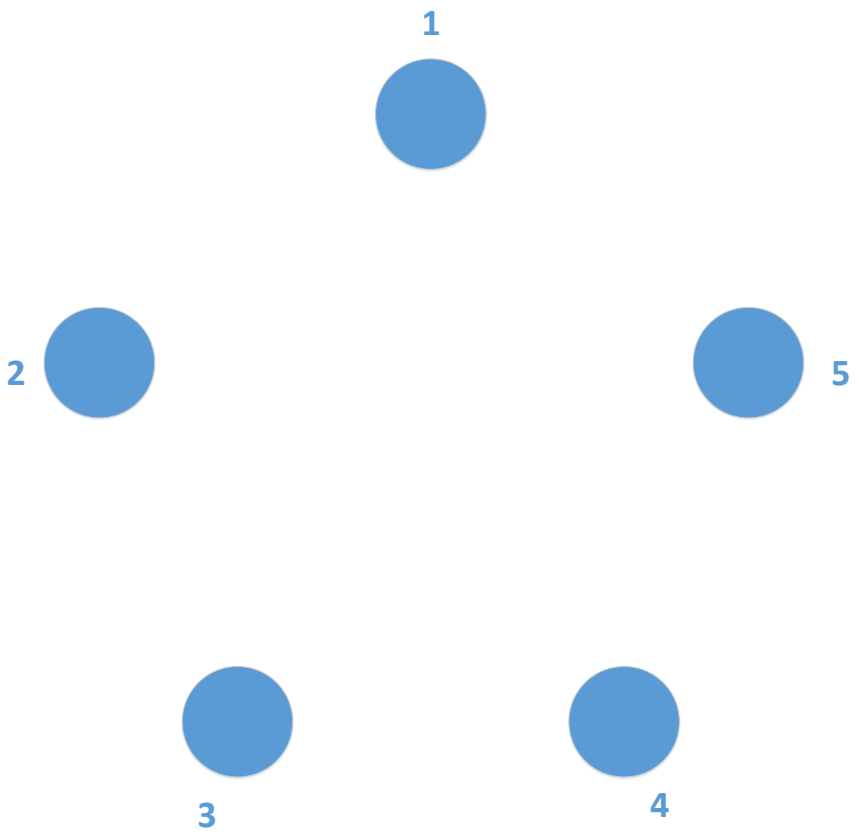
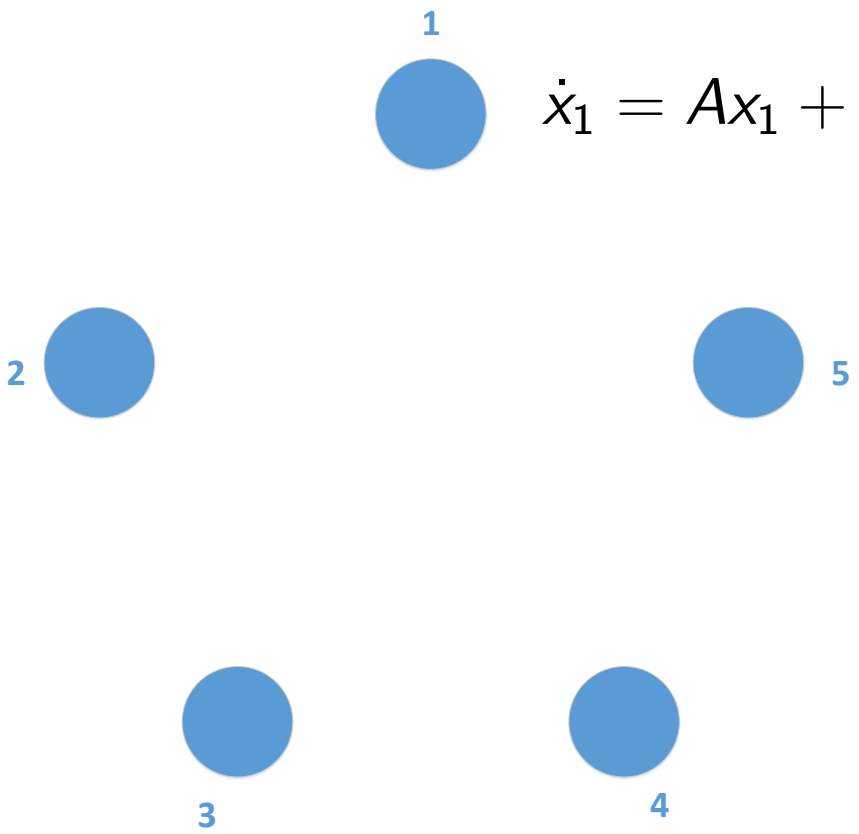
A Simple Example

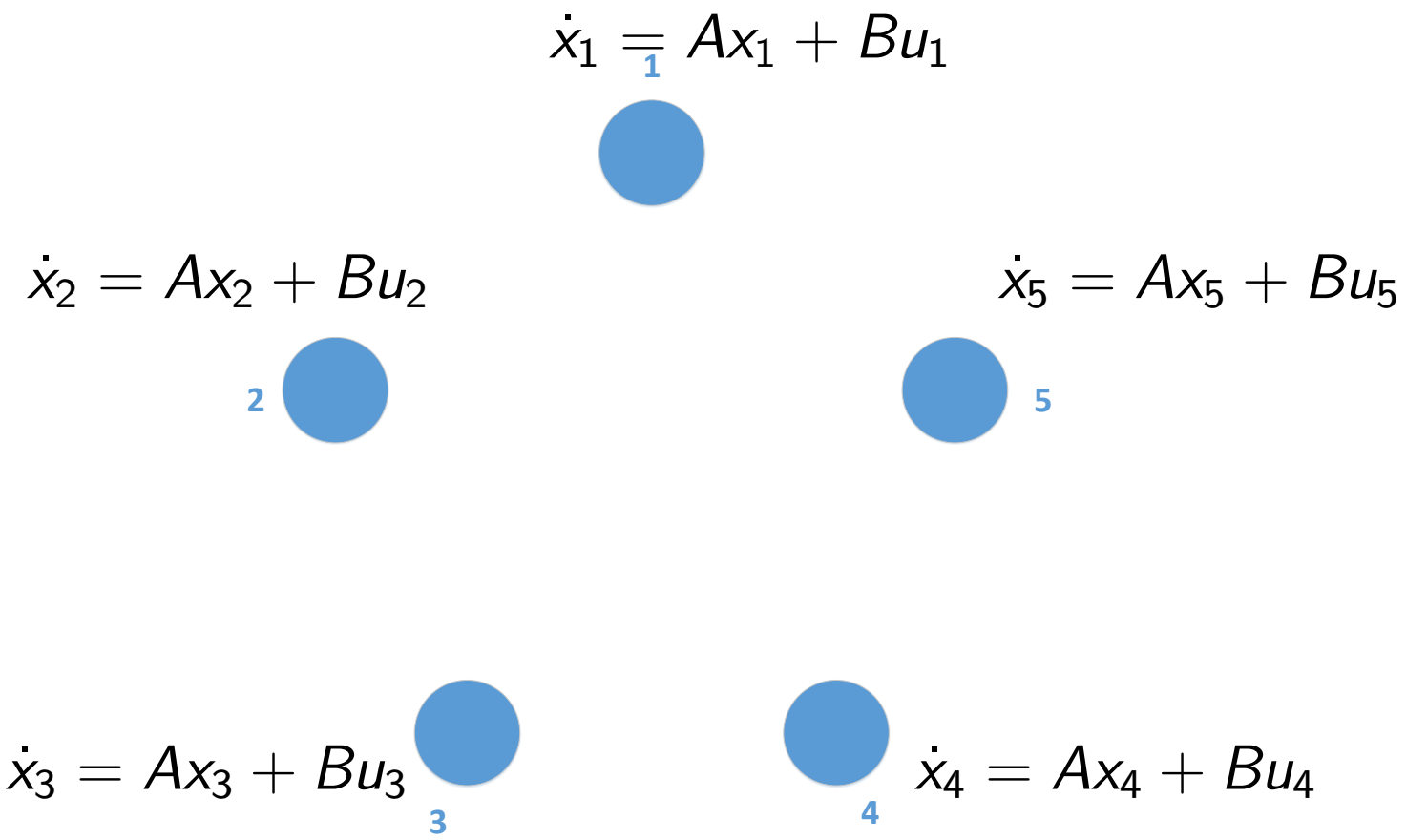


A Simple Example



$$\dot{x}_1 = Ax_1 + Bu_1$$

A Simple Example

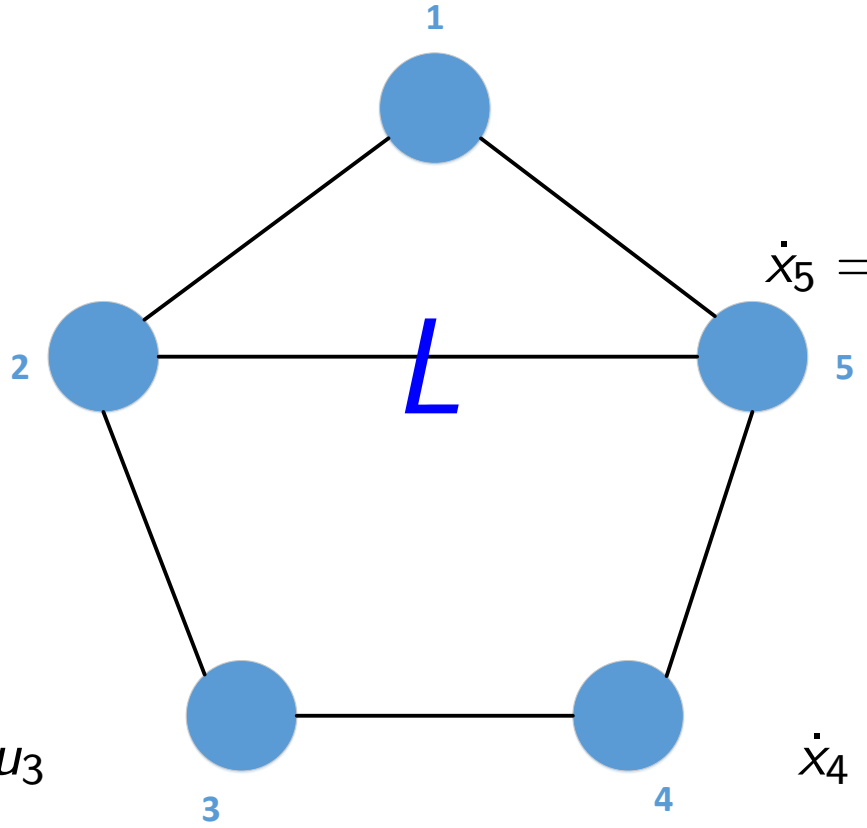


A Simple Example

$$\dot{x}_1 = Ax_1 + Bu_1$$

$$\dot{x}_2 = Ax_2 + Bu_2$$

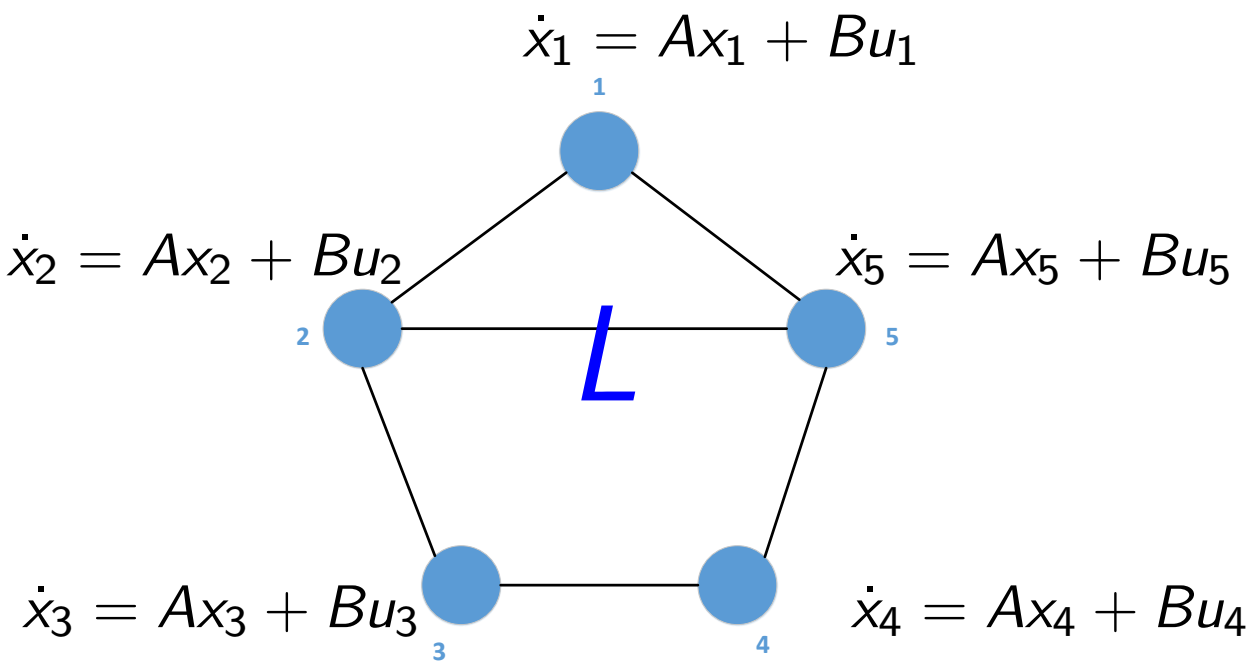
$$\dot{x}_5 = Ax_5 + Bu_5$$



$$\dot{x}_3 = Ax_3 + Bu_3$$

$$\dot{x}_4 = Ax_4 + Bu_4$$

A Simple Example

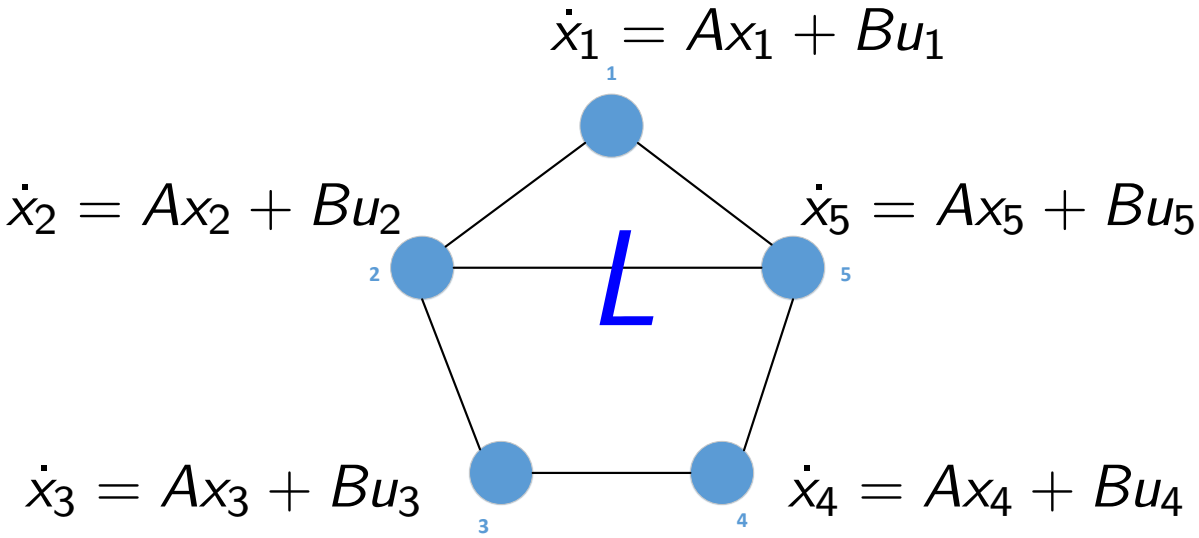


Find a distributed control law

$$u_i = \sum_{j \in \mathcal{N}_i} K(x_j - x_i) \qquad \text{or} \qquad u = (L \otimes K)x$$

such that the states of the agents reach consensus: $\lim_{t\to\infty}(x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, 5$.

A Simple Example



Consider a linear quadratic cost functional ($Q \geq 0, R > 0$):

$$J(u) = \int_0^\infty \frac{1}{2} \sum_{i=1}^{5} \sum_{j \in \mathcal{N}_i} (x_i - x_j)^\top Q (x_i - x_j) + \sum_{i=1}^{5} u_i^\top R u_i \; dt, \;\; Q \geq 0, R > 0$$

Find a control law $u_i = \sum_{j \in \mathcal{N}_i} K(x_j - x_i)$ such that $J(u)$ is minimized and $\lim_{t \to \infty} (x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, 5$.

Distributed LQ Problem

Consider a group of N identical agents and undirected communication topology.

- **Agent Dynamics:**

$$\dot{x}_i(t) = Ax_i(t) + Bu_i(t), \quad x_i(0) = x_{i0}, \quad i = 1, 2, \ldots, N$$

- **Cost functional:**

$$J(u) = \int_0^\infty \frac{1}{2} \sum_{i=1}^{N} \sum_{j \in \mathcal{N}_i} (x_i - x_j)^\top Q (x_i - x_j) + \sum_{i=1}^{N} u_i^\top R u_i \, dt, \; Q \geq 0, R > 0. \tag{1}$$

- **Distributed controller:**

$$u_i = \sum_{j \in \mathcal{N}_i} K(x_j - x_i), \quad i = 1, 2, \ldots, N$$

- **Aim:** Find K such that
  - $\lim_{t \to \infty} (x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, N$.
  - $J(K)$ is minimized.

Distributed LQ Problem

Denote $x = \left(x_1^\top, \ldots, x_N^\top\right)^\top$, $u = \left(u_1^\top, \ldots, u_N^\top\right)^\top$.

- **Agents:**
$$\dot{x} = (I_N \otimes A)x + (I_N \otimes B)u, \quad x(0) = x_0$$
- **Communication:** undirected connected graph with Laplacian L.
- **Cost functional:**
$$J(u) = \int_0^\infty x^\top (L \otimes Q)x + u^\top (I_N \otimes R)u \; dt, \quad Q \geq 0, R > 0.$$
- **Distributed controller:** $\quad u = (L \otimes K)x$
- **Aim:** Find K such that
  - $\lim_{t\to\infty}(x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, N$.
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Distributed LQ Problem

Denote $x = \left(x_1^\top, \ldots, x_N^\top\right)^\top$, $u = \left(u_1^\top, \ldots, u_N^\top\right)^\top$.

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- **Distributed controller:** $\quad u = (L \otimes K)x$
- **Aim:** Find K such that
  - $\lim_{t\to\infty}(x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, N$.
  - $J(K)$ is minimized.

However, this problem is non-convex and it is difficult to solve in general. Therefore, we consider a suboptimal version of this problem.

Distributed Suboptimal LQ Problem

- **Agents:**
$$\dot{x} = (I_N \otimes A)x + (I_N \otimes B)u, \quad x(0) = x_0$$
- **Cost functional:**
$$J(u) = \int_0^\infty x^\top (L \otimes Q)x + u^\top (I_N \otimes R)u \; dt, \quad Q \geq 0, R > 0.$$
- **Distributed controller:** $\quad u = (L \otimes K)x$
- **Aim:** Find K such that
  - $\lim_{t\to\infty}(x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, N$.
  - Given $\gamma > 0$. $J(K) < \gamma$.

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Variable Transformation

- **Distributed controller:** $u = (L \otimes K)x$
- **Controlled network:**

$$\dot{x} = (I_N \otimes A + L \otimes BK)x, \quad x(0) = x_0$$

- **Cost functional:**

$$J(u) = \int_0^\infty x^\top (L \otimes Q)x + u^\top (I_N \otimes R)u \; dt, \quad Q \geq 0, R > 0.$$

- **Aim:**
  - $\lim_{t\to\infty}(x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, N$.
  - $J(K) < \gamma$.

Note that $U^\top L U = \Lambda = \mathrm{diag}(0, \lambda_2, \ldots, \lambda_N)$, therefore our first attempt is to apply variable transformations

$$\bar{x} = (U^\top \otimes I_n)x, \quad \bar{u} = (U^\top \otimes I_m)u.$$

Variable Transformation

By applying the following variable transformations:

$$\bar{x} = (U^\top \otimes I_n)x, \quad \bar{u} = (U^\top \otimes I_m)u,$$

The controlled network becomes

$$\dot{\bar{x}} = (I_N \otimes A + \Lambda \otimes BK)\,\bar{x}. \tag{2}$$

The cost $J(K)$ is now

$$J(K) = \int_0^\infty \sum_{i=1}^{N} \bar{x}_i^\top (\lambda_i Q + \lambda_i^2 K^\top R K)\bar{x}_i \; dt. \tag{3}$$

Note that the two equations above can be written in a decoupled way.

Variable Transformation

We split equation (2) into decoupled equations

$$\dot{\bar{x}}_1 = A\bar{x}_1, \tag{4}$$

$$\dot{\bar{x}}_i = (A + \lambda_i BK)\bar{x}_i, \quad i = 2, 3, \ldots, N, \tag{5}$$

together with the cost

$$J(K) = \sum_{i=2}^{N} J_i(K), \quad J_i(K) = \int_0^\infty \bar{x}_i^\top (\lambda_i Q + \lambda_i^2 K^\top RK)\bar{x}_i \; dt \tag{6}$$

We can assign $J_i(K)$ to (5) for $i = 2, 3, \ldots, N$.

Before we go further, we first state a well-know(in network systems) result.

A Useful Lemma

Lemma 1

The controlled network

$$\dot{\bar{x}} = (I_N \otimes A + \Lambda \otimes BK)\,\bar{x}$$

reaches consensus under the distributed control law

$$u = (L \otimes K)x$$

if and only if, for $i = 2, 3, \ldots, N$*, systems*

$$\dot{\bar{x}}_i = (A + \lambda_i BK)\bar{x}_i,$$

are stable.

Transformed Problem

Therefore, the problem is now to find controllers $u = \lambda_i K x$, $i = 2, 3, \ldots, N$ such that systems

$$\dot{\bar{x}}_i = A\bar{x}_i + B\bar{u}_i, \quad i = 2, 3, \ldots, N \tag{7}$$

are stable and the cost

$$J(K) = \sum_{i=2}^{N} J_i(K), \quad J_i(K) = \int_0^\infty \bar{x}_i^\top (\lambda_i Q + \lambda_i^2 K^\top R K) \bar{x}_i \, dt \tag{8}$$

satisfies $J(K) < \gamma$.

We then have the following lemma.

Lemma

Lemma 2

Consider the systems (7) with associated cost functionals $J_i(K)$, $i = 2, 3, \ldots, N$, respectively. Let $\gamma > 0$. Then all systems

$$\dot{\bar{x}}_i = A\bar{x}_i + B\bar{u}_i, \quad i = 2, 3, \ldots, N,$$

are stable and $J(K) < \gamma$ if and only if there exist positive semi-definite matrices P_i satisfying

$$(A + \lambda_i BK)^\top P_i + P_i(A + \lambda_i BK) + \lambda_i Q + \lambda_i^2 K^\top RK < 0,$$

$$\sum_{i=2}^{N} \bar{x}_{i0}^\top P_i \bar{x}_{i0} < \gamma,$$

for $i = 2, 3, \ldots, N$, respectively.

Discussion

Lemma 2 establishes a *necessary* and *sufficient* condition for control gain K to stabilize all systems

$$\dot{\bar{x}}_i = A\bar{x}_i + B\bar{u}_i, \quad i = 2, 3, \ldots, N,$$

and to satisfy $J(K) < \gamma$.

- However, Lemma 2 does yet not provide a method to compute such a control gain K.
- Moreover, we want to solve ONE instead of $N - 1$ Riccati inequalities to compute such control gain K.

To this end, we further provide two design methods to compute such K and, correspondingly, two suboptimal distributed control laws.

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Distributed Suboptimal LQ Problem

- **Agents:**

$$\dot{x} = (I_N \otimes A)x + (I_N \otimes B)u, \quad x(0) = x_0 \tag{9}$$

- **Cost functional:**

$$J(u) = \int_0^\infty x^\top (L \otimes Q)x + u^\top (I_N \otimes R)u \; dt, \quad Q \geq 0, R > 0. \tag{10}$$

- **Communication topology:** undirected connected graph with Laplacian L.
- **Distributed controller:**

$$u = (L \otimes K)x \tag{11}$$

- **Aim:** Find K such that
  - $\lim_{t\to\infty}(x_i - x_j) \to 0$ for $i, j = 1, 2, \ldots, N$.
  - Given $\gamma > 0$. $J(K) < \gamma$.

Theorem 3

Consider multi-agent system (9) with associated cost functional (10). Assume that the underlying graph is undirected and connected. Let $\gamma > 0$. Choose c such that

$$\frac{2}{\lambda_2 + \lambda_N} \leq c < \frac{2}{\lambda_N}.$$

Then there exists a positive semi-definite P satisfying the Riccati inequality

$$A^\top P + PA + (c^2\lambda_N^2 - 2c\lambda_N)PBR^{-1}B^\top P + \lambda_N Q < 0.$$

Assume, moreover, that P can be found such that

$$x_0^\top \left(\left(I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top \right) \otimes P \right) x_0 < \gamma.$$

Let $K := -cR^{-1}B^\top P$. Then the controlled network reaches consensus and the control law (11) is suboptimal, i.e., $J(K) < \gamma$.

Theorem 4

Consider multi-agent system (9) with associated cost functional (10). Assume that the underlying graph is undirected and connected. Let $\gamma > 0$. Choose c such that

$$0 < c < \frac{2}{\lambda_2 + \lambda_N}.$$

Then there exists a positive semi-definite P satisfying Riccati inequality

$$A^\top P + PA + (c^2\lambda_2^2 - 2c\lambda_2)PBR^{-1}B^\top P + \lambda_N Q < 0.$$

Assume, moreover, that P can be found such that

$$x_0^\top \left(\left(I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top \right) \otimes P \right) x_0 < \gamma. \tag{12}$$

Let $K := -cR^{-1}B^\top P$. Then the controlled network (14) reaches consensus and the control law (11) is suboptimal, i.e., $J(K) < \gamma$.

Theorem 5

Let a lower bound for λ_2 be given by $l_2 > 0$ and an upper bound for λ_N be given by L_N. Let $\gamma > 0$. Choose c such that $\frac{2}{l_2+L_N} \leq c < \frac{2}{L_N}$.Define $K = -cR^{-1}B^\top P$, with $P \geq 0$ satisfying

$$A^\top P + PA + (c^2 L_N^2 - 2cL_N)PBR^{-1}B^\top P + L_N Q < 0. \tag{13}$$

Then the controlled network reaches consensus and the distributed control law $L \otimes K$ is suboptimal, i.e. $J(K) < \gamma$, for all initial states x_0 that satisfy (12).
Furthermore, if we choose c such that $0 < c < \frac{2}{l_2+L_N}$.and define $K = -cR^{-1}B^\top P$ with $P \geq 0$ satisfying

$$A^\top P + PA + (c^2 l_2^2 - 2cl_2)PBR^{-1}B^\top P + L_N Q < 0, \tag{14}$$

then, still, the controlled network reaches consensus and the distributed control law $L \otimes K$ is suboptimal for all x_0 that satisfy (12).

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Summary

- We formulate a distributed suboptimal linear quadratic control problem.
- Then, we provide two design methods to compute such suboptimal distributed controllers, which guarantee the controlled network to reach consensus and the associated cost to be smaller than an a priori given upper bound.
- Furthermore, we relax the requirement of exact knowledge of the smallest nonzero and largest eigenvalue of the graph Laplacian by using only lower and upper bounds.

Thank you!